1. Objective

The objective of this lab is to build a multi-router network in Cisco Packet Tracer, assign IPv4 addresses with different subnet masks to each router interface, enable OSPF (Open Shortest Path First) so that all internal routers learn each other's networks, and configure NAT with overload (PAT) so that hosts on the internal 192.168.1.0/24 network can reach an external host through a pool of public addresses.

- Configure hostnames and IP addressing on five routers (R1, R2, R3, NAT/PAT and ISP).
- Configure OSPF process 1 in area 0 with unique router IDs.
- Configure NAT/PAT using an address pool, an access list and the overload keyword.
- Verify end-to-end connectivity with ping and inspect the NAT translation table.

2. Equipment and Tools

- Cisco Packet Tracer network simulator
- Five Cisco routers with GigabitEthernet interfaces: R1, R2, R3, NAT/PAT and ISP
- End devices used for console access and connectivity tests (PC in the 192.168.1.0/24 LAN and a remote host at 7.0.0.10)

3. Network Design and Addressing

R1 connects the internal LAN (192.168.1.0/24) and has links to R2 and R3. R2 and R3 both connect to the NAT/PAT router, forming a redundant path. The NAT/PAT router connects to the ISP router, which serves the external network 7.0.0.0/25. The addressing used in this lab is summarized below.

Router	Interface	IP Address	Subnet Mask	Connected To
R1	G0/0	192.168.1.1	255.255.255.0 (/24)	Internal LAN
R1	G0/1	192.168.2.1	255.255.255.128 (/25)	R2 G0/0
R1	G0/2	192.168.5.1	255.255.255.0 (/24)	R3 G0/0
R2	G0/0	192.168.2.2	255.255.255.128 (/25)	R1 G0/1
R2	G0/1	192.168.3.1	255.255.255.192 (/26)	NAT/PAT G0/0
R3	G0/0	192.168.5.2	255.255.255.0 (/24)	R1 G0/2
R3	G0/1	192.168.4.1	255.255.255.224 (/27)	NAT/PAT G0/1
NAT/PAT	G0/0	192.168.3.2	255.255.255.192 (/26)	R2 G0/1
NAT/PAT	G0/1	192.168.4.2	255.255.255.224 (/27)	R3 G0/1
NAT/PAT	G0/2	10.0.0.1	255.255.255.0 (/24)	ISP G0/0 (outside)
ISP	G0/0	10.0.0.2	255.255.255.0 (/24)	NAT/PAT G0/2
ISP	G0/1	7.0.0.1	255.255.255.128 (/25)	External network

Table 1: IP addressing scheme

4. Procedure

Part A: Basic Router Configuration (Hostname and IP Addressing)

On each router, privileged EXEC mode was entered with enable, global configuration mode with configure terminal, and the hostname was set. Each interface was then given an IP address and subnet mask and enabled with no shutdown. The configuration was saved with do wr. The interfaces came up and the line protocol changed to up, confirming the physical links were working.

R1 was configured with interfaces g0/0 (192.168.1.1/24), g0/1 (192.168.2.1/25) and g0/2 (192.168.5.1/24).

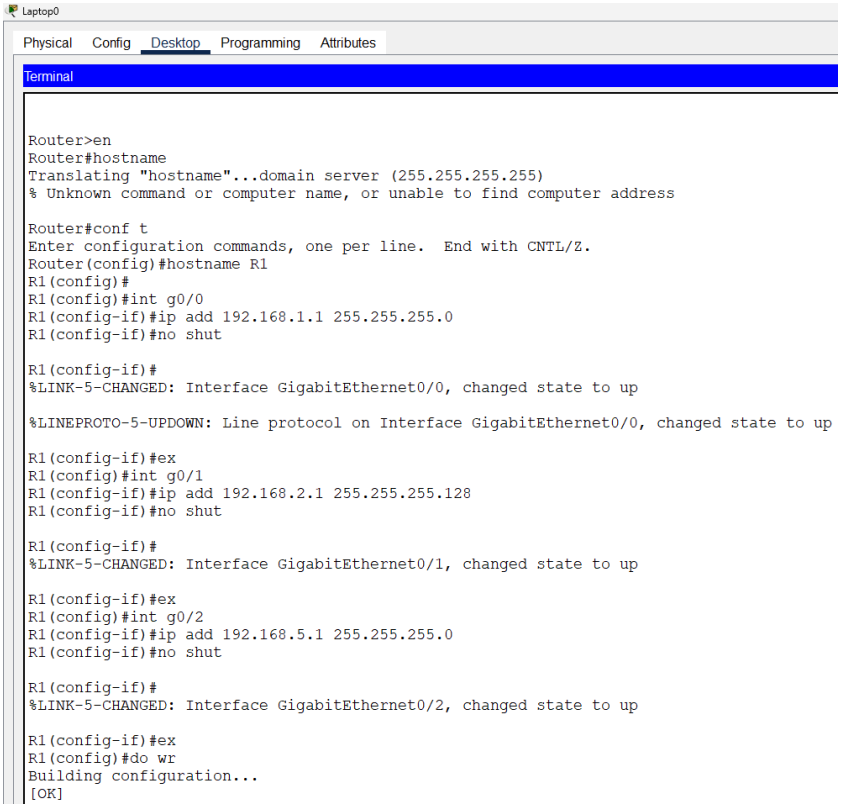


Figure 1: *R1 hostname and interface configuration*

R2 was configured with g0/0 (192.168.2.2/25) toward R1 and g0/1 (192.168.3.1/26) toward the NAT/PAT router.

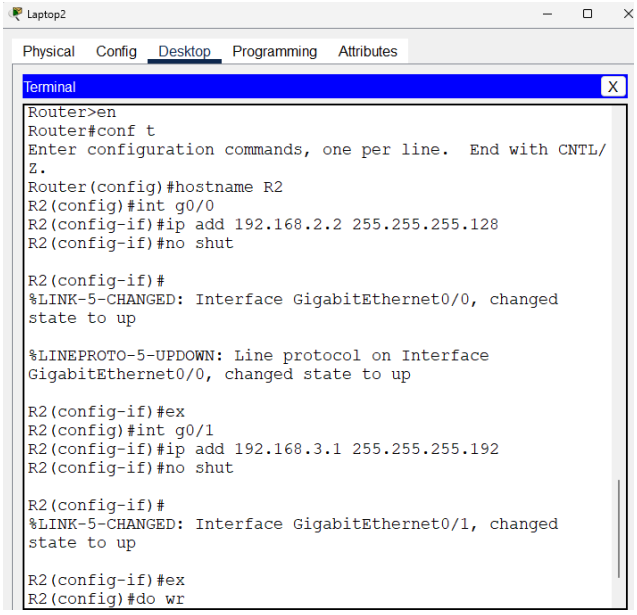


Figure 2: *R2 hostname and interface configuration*

R3 was configured with g0/0 (192.168.5.2/24) toward R1 and g0/1 (192.168.4.1/27) toward the NAT/PAT router. A mistyped command (ap add) was rejected with an invalid input error and re-entered correctly.

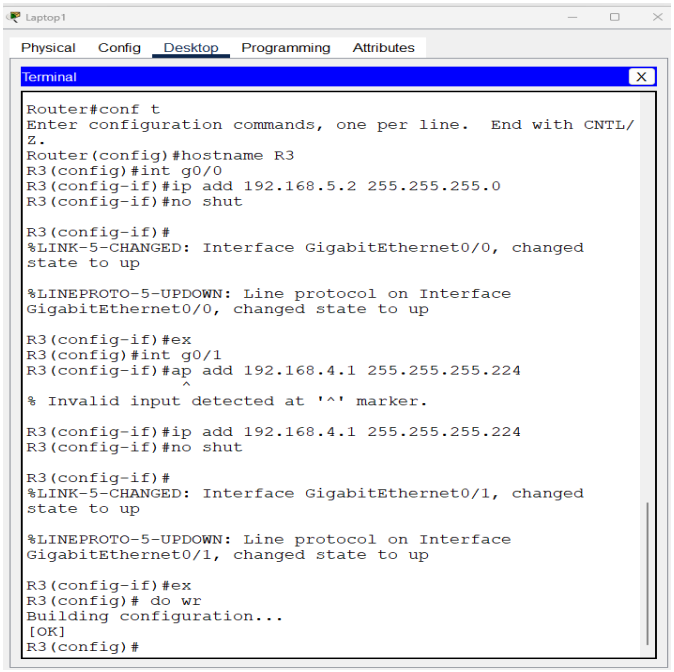


Figure 3: *R3 hostname and interface configuration (including a rejected mistyped command)*

The NAT/PAT router was configured with g0/0 (192.168.3.2/26), g0/1 (192.168.4.2/27) and g0/2 (10.0.0.1/24). An incomplete interface command (int g0/) was rejected and then re-entered.

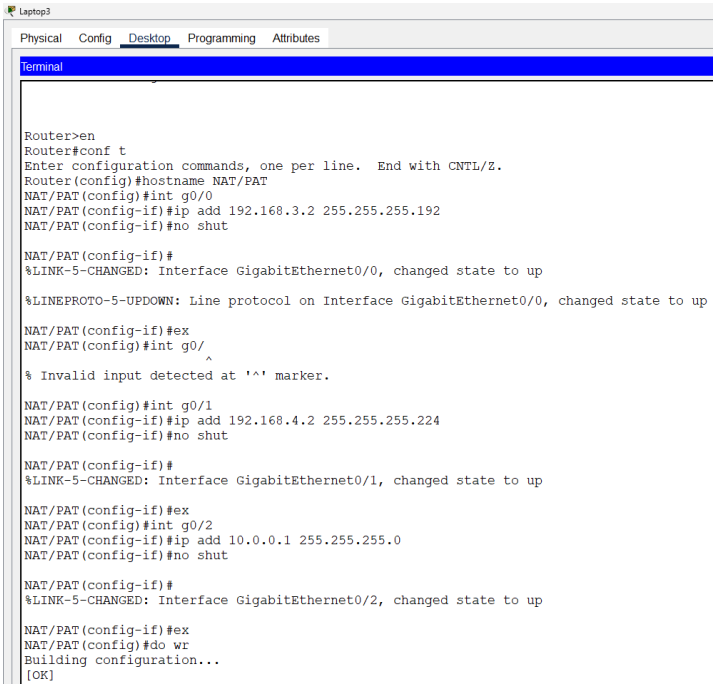


Figure 4: *NAT/PAT router hostname and interface configuration*

The ISP router was configured with g0/0 (10.0.0.2/24) and g0/1 (7.0.0.1). The first mask entered for g0/1 (255.255.255.108) was rejected as a bad mask, so the valid mask 255.255.255.128 was used instead.

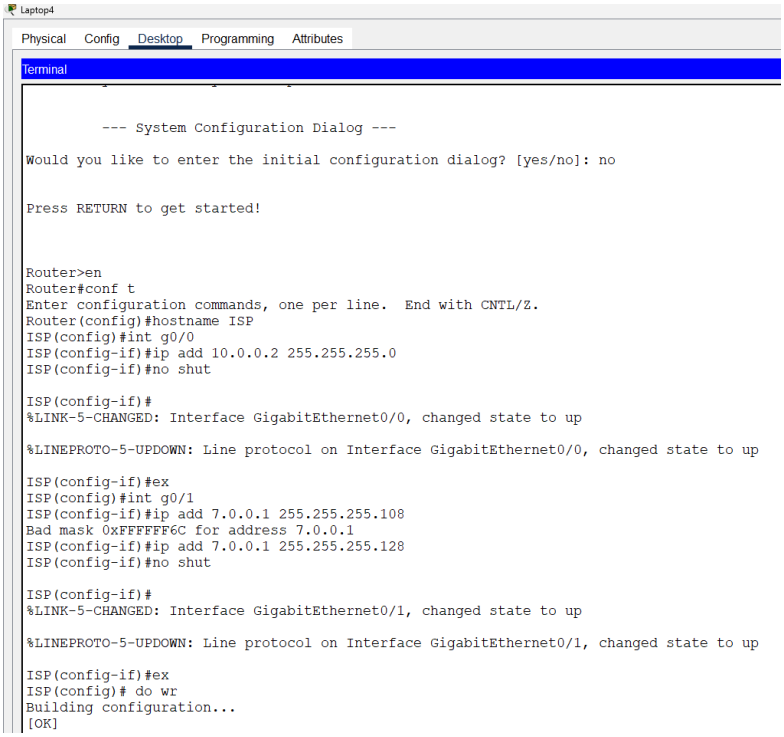


Figure 5: *ISP router hostname and interface configuration (bad mask corrected)*

Part B: OSPF Configuration

OSPF process 1 was enabled on R1, R2, R3 and the NAT/PAT router. Each router was given a unique router ID, and each connected network was advertised into area 0 using the network command with the appropriate wildcard mask. Wildcard masks match the subnet masks used earlier: 0.0.0.255 for /24, 0.0.0.127 for /25, 0.0.0.63 for /26 and 0.0.0.31 for /27.

Router	Router ID	Networks Advertised (area 0)
R1	1.1.1.1	192.168.1.0 0.0.0.255, 192.168.2.0 0.0.0.127, 192.168.5.0 0.0.0.255
R2	2.2.2.2	192.168.2.0 0.0.0.127, 192.168.3.0 0.0.0.63
R3	3.3.3.3	192.168.5.0 0.0.0.255, 192.168.4.0 0.0.0.31
NAT/PAT	4.4.4.4	192.168.3.0 0.0.0.63, 192.168.4.0 0.0.0.31, 10.0.0.0 0.0.0.255

Table 2: OSPF configuration summary

R1 advertised its three connected networks in area 0.

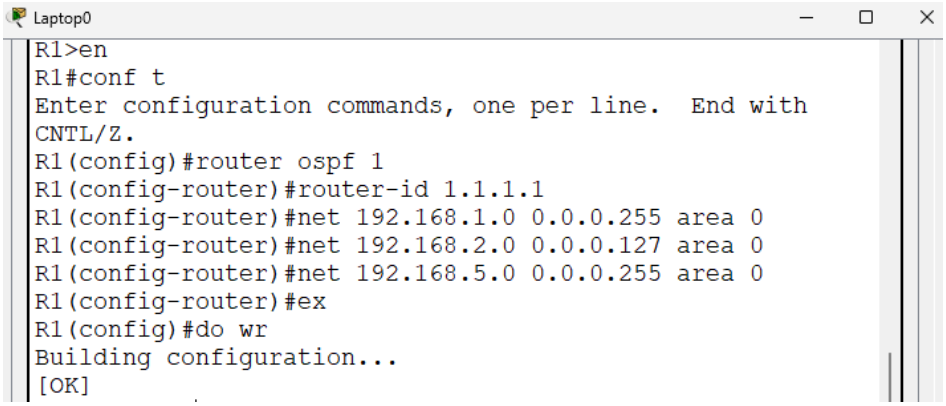


Figure 6: *OSPF configuration on R1*

R2 advertised 192.168.2.0/25 and 192.168.3.0/26. The first attempt at the second network command was incomplete and was rejected, then re-entered with the correct wildcard mask. The adjacency message shows R2 forming a neighbor relationship with R1 (1.1.1.1).

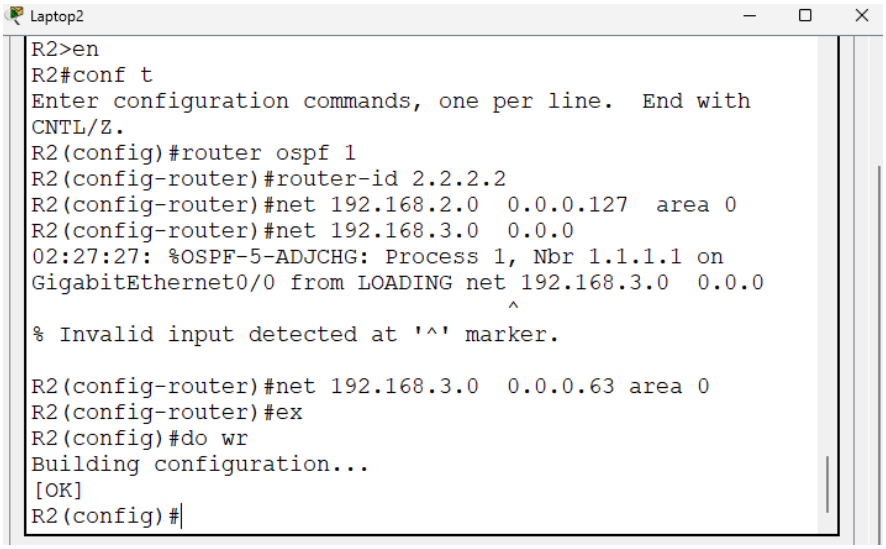


Figure 7: *OSPF configuration on R2 showing adjacency with R1*

R3 advertised 192.168.5.0/24 and 192.168.4.0/27. After an incomplete command was rejected it was re-entered with wildcard 0.0.0.31.

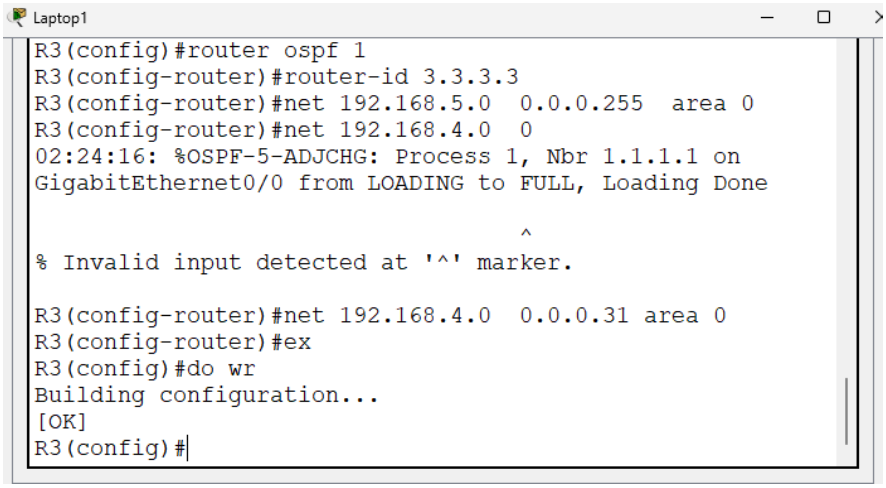


Figure 8: *OSPF configuration on R3 showing adjacency with R1*

The NAT/PAT router advertised its links toward R2, R3 and the ISP. The log messages show OSPF adjacencies reaching the FULL state with R2 (2.2.2.2) and R3 (3.3.3.3).

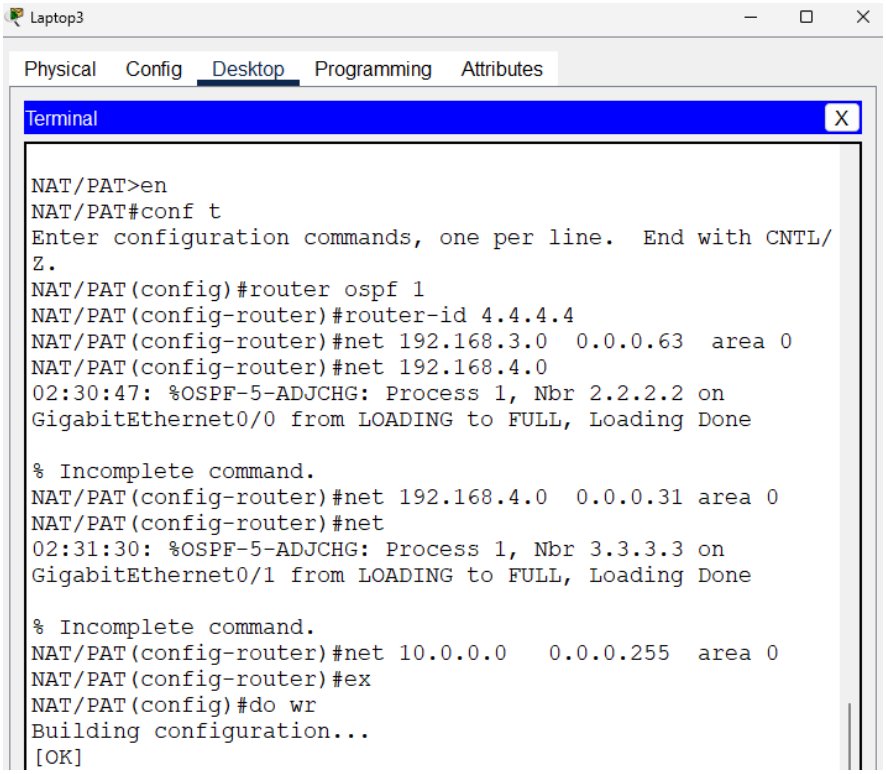


Figure 9: *OSPF configuration on the NAT/PAT router showing adjacencies with R2 and R3*

Part C: NAT/PAT (Overload) Configuration

NAT with overload was configured so that internal hosts share a small pool of public addresses. A standard access list (access-list 1) permits the internal LAN 192.168.1.0/24. A NAT pool named nPool was created with the range 100.100.100.1 to 100.100.100.6 (mask 255.255.255.248). Interfaces facing the internal network were marked ip nat inside and the interface toward the ISP (g0/2) was marked ip nat outside. Finally, ip nat inside source list 1 pool nPool overload was applied.

The NAT/PAT router configuration is shown below. An attempt to remove a physical interface with no int g0/1 was refused by the router, and the interface was then simply re-selected and configured.

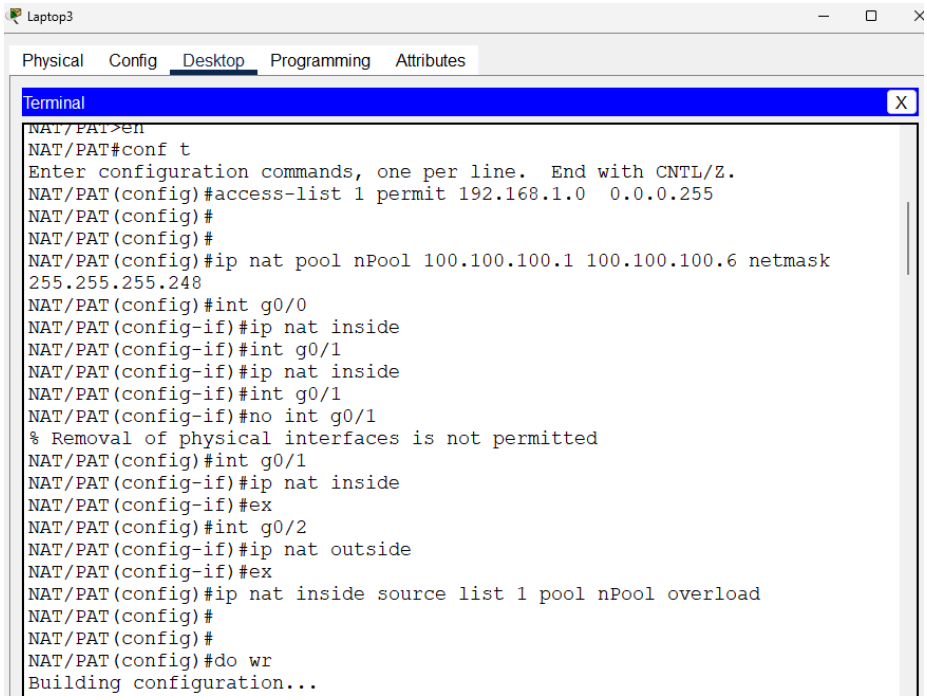


Figure 10: *NAT/PAT router: access list, NAT pool, inside/outside interfaces and overload command*

The ISP router was also configured with the same pool name and range, with g0/0 and g0/1 marked as NAT inside, g0/2 marked as NAT outside, and the overload statement applied.

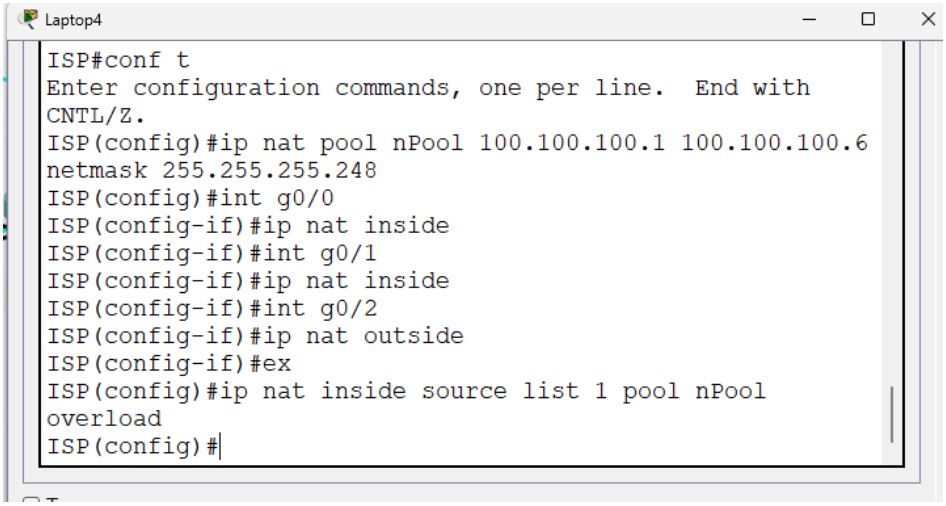


Figure 11: *NAT pool and NAT interface configuration on the ISP router*

Part D: Verification

Connectivity was tested by pinging the external host 7.0.0.10 from the internal host. All four echo requests received replies with 0% packet loss and a TTL of 124, confirming that OSPF routing and NAT are working end to end.

```
C:\>
C:\>ping 7.0.0.10

Pinging 7.0.0.10 with 32 bytes of data:

Reply from 7.0.0.10: bytes=32 time<1ms TTL=124
Reply from 7.0.0.10: bytes=32 time<1ms TTL=124
Reply from 7.0.0.10: bytes=32 time<1ms TTL=124
Reply from 7.0.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 7.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 12: *Successful ping from the internal host to 7.0.0.10 (0% loss)*

The NAT table was cleared with clear ip nat translation * and the ping was repeated. The show ip nat translations output on the NAT/PAT router lists four ICMP entries. The inside local address 192.168.1.10 was translated to the inside global address 100.100.100.1, while the outside address remained 7.0.0.10. This confirms that PAT translated the private source address to a public pool address.

```
NAT/PAT#clear ip nat translation *
NAT/PAT#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 100.100.100.1:21  192.168.1.10:21    7.0.0.10:21        7.0.0.10:21
icmp 100.100.100.1:22  192.168.1.10:22    7.0.0.10:22        7.0.0.10:22
icmp 100.100.100.1:23  192.168.1.10:23    7.0.0.10:23        7.0.0.10:23
icmp 100.100.100.1:24  192.168.1.10:24    7.0.0.10:24        7.0.0.10:24

NAT/PAT#
```

Figure 13: *NAT translation table on the NAT/PAT router*

5. Results and Observations

- All router interfaces came up and the line protocol changed to up after configuration.
- OSPF adjacencies formed between R1 and R2, R1 and R3, R2 and the NAT/PAT router, and R3 and the NAT/PAT router, reaching the FULL state.
- The ping from 192.168.1.10 to 7.0.0.10 succeeded with 0% loss.
- The NAT table showed the private address 192.168.1.10 translated to 100.100.100.1, with a different port or ICMP identifier for each request, demonstrating PAT (overload).
- Several typing errors (invalid input, incomplete command, bad mask) occurred during configuration. Each was reported by the IOS command parser and fixed by re-entering the command correctly.
- The screenshots do not include the routing table or ISP routing configuration, so the return path from the ISP to the NAT pool is not documented here.

6. Conclusion

In this lab, a five-router topology was configured in Cisco Packet Tracer using several different subnet sizes (/24, /25, /26 and /27). OSPF in area 0 allowed the internal routers to exchange routes automatically, and NAT with overload on the edge router translated the internal private network to a pool of public addresses. The successful ping and the NAT translation table verified the configuration. The lab reinforced subnet and wildcard mask calculation, OSPF neighbor formation, the role of inside and outside NAT interfaces, and the importance of reading IOS error messages when troubleshooting.